

46 YEARS OF EXCELLENCE | NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **SPCL-GTM-02 (AN)** **Date: 12.01.2025**
Time: 3.00Hrs. **JEE MAIN MODEL** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A soap bubble is blown to a diameter of 7 cm. 36960 erg of work is done in blowing it further. If surface tension of soap solution is 40 dyne/cm then the new radius is____cm Take $\left(\pi = \frac{22}{7}\right)$.

   A) 21 B) 28 C) 7 D) 14

2. An infinite plane sheet of charge having uniform surface charge density $+\sigma_s C/m^2$ is placed in x-y plane. Another infinitely long line charge having uniform charge density

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$+\lambda_e C/m$ is placed at z = 4m plane and parallel to y-axis. If the magnitude value $|\sigma_s| = 2|\lambda_e|$ then at point (0, 0, 2), the ratio of magnitude of electric field values due to sheet charge to that of line charge is $\pi\sqrt{n}:1$. The value of n is

A) 8 B) 16 C) 24 D) 32

3. A body travels 102.5m in nth second and 115.0m in $(n+2)^{th}$ second. The acceleration is :

A) $12.5\,m/s^2$ B) $5\,m/s^2$ C) $6.25\,m/s^2$ D) $9\,m/s^2$

4. Which of the following nuclear fragments corresponding to nuclear fission between neutron $\left(\begin{matrix}1\\0\end{matrix}n\right)$ and uranium isotope $\left(\begin{matrix}235\\92\end{matrix}U\right)$ is correct :

A) ${}^{153}_{51}Sb + {}^{99}_{41}Nb + 3{}^{1}_{0}n$ B) ${}^{144}_{56}Ba + {}^{89}_{36}Kr + 3{}^{1}_{0}n$

C) ${}^{144}_{56}Ba + {}^{89}_{36}Kr + 4{}^{1}_{0}n$ D) ${}^{140}_{56}Xe + {}^{94}_{38}Sr + 3{}^{1}_{0}n$

5. Five identical convex lenses are placed one after the other in close contact. The power of this arrangement is 25D. Then, the focal length of each lens….

A) 25 cm B) 500 cm C) 20 cm D) 50 cm

6. The resistance of the platinum wire of a platinum resistance thermometer at the ice point and steam point are 8Ω and 10Ω respectively. After inserting in a hot bath of temperature 400^0C, the resistance of platinum wire is :

A) 16Ω B) 2Ω C) 8Ω D) 10Ω

7. The equation of stationary wave is given as $y = 2A\sin\left(\frac{2\pi}{\lambda}nt\right)\cos\left(\frac{2\pi}{\lambda}x\right)$, then which of the following is not correct.

A) Dimension of n is $\left[LT^{-1}\right]$ B)Dimension of nt is [L]

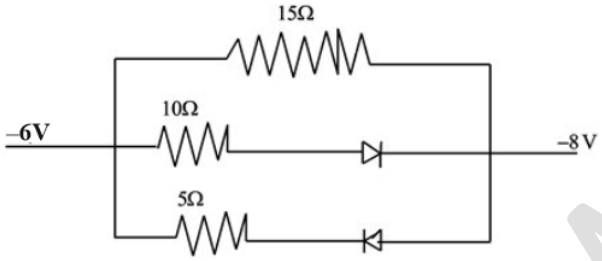
C) Dimension of x is [L] D) Dimension of $\frac{n}{\lambda}$ is [T]

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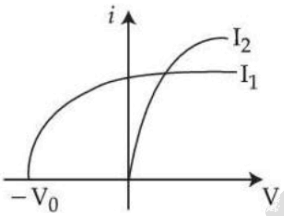
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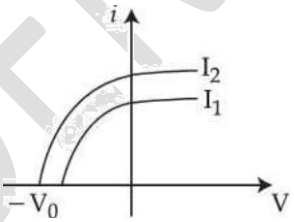
8. The value of net resistance of the network as shown in the given figure is :

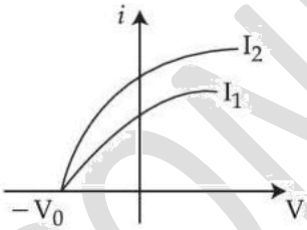


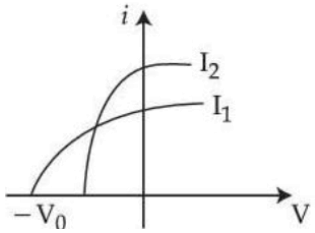
A) $(5/2)\Omega$ B) 6Ω C) $(30/11)\Omega$ D) $(15/4)\Omega$

9. Which figure shows the correct variation of applied potential difference (V) with photoelectric current (I) at two different intensities of light ($I_1 < I_2$) of same wavelengths:

A) 

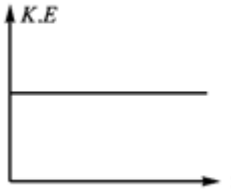
B) 

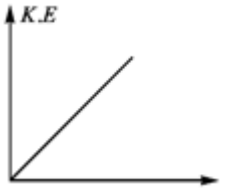
C) 

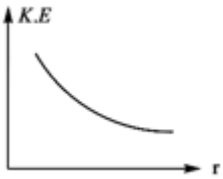
D) 

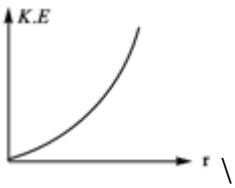
10. An infinitely long positively charged straight thread has a linear charge density λCm^{-1}. An electron revolves along a circular path having axis along the length of the wire. The graph that correctly represents the variation of the kinetic energy of electron as a function of radius of circular path from the wire is:

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A) 

B) 

C) 

D) 

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11. An electron is projected with uniform velocity along the axis inside a current carrying long solenoid, then

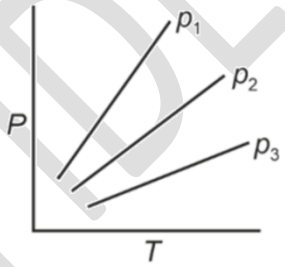
A) The electron will continue to move with uniform velocity along the axis of the solenoid

B) The electron will experience a force at 45^0 to the axis and execute a helical path.

C) The electron path will be circular about the axis

D) The electron will be accelerated along the axis

12. P-T diagram of an ideal gas having three different densities ρ_1, ρ_2 and ρ_3 (in three different cases) is shown in the figure. Which of the following is correct:



A) $\rho_3 = \rho_2 = \rho_1$ B) $\rho_1 > \rho_2$ C) $\rho_1 < \rho_2$ D) $\rho_2 < \rho_3$

13. A metal wire of uniform mass density having length L and mass M is bent to form a semicircular arc and a particle of mass m is placed at the centre of the arc. The gravitational force on the particle by the wire is :

A)0 B) $\frac{GMm\pi^2}{L^2}$ C) $\frac{2GMm\pi}{L^2}$ D) $\frac{GMm\pi}{2L^2}$

14. Given below are two statements:

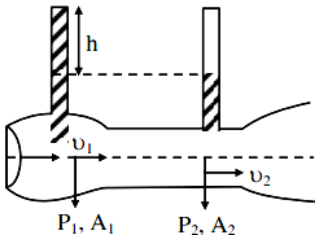
Statements I: When speed of liquid is zero everywhere, pressure difference at any two points depends on equation $P_1 - P_2 = \rho g(h_2 - h_1)$.

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Statement II : In ventury tube shown $2gh = v_1^2 - v_2^2$



In the light of the above statements, choose the most appropriate answer from the options given below.

A)Both Statement I and Statement II are correct

B) Statement I is incorrect but Statement II is correct.

C) Both Statement I and Statement II are incorrect

D) Statement I is correct but Statement II is incorrect.

15. In an ac circuit , the instantaneous current is zero, when the instantaneous voltage is maximum. In this case , the source may be connected to :

a) Pure inductor b) Pure capacitor c) pure resistor

d) Combination of an inductor and capacitor

Choose the correct answer from the options given below :

A) b, c and d only B) a and b only C) a,b,c only D) a,b,d only

16. In an experiment to measure focal length (f) of convex lens, the least counts of the measuring scales for the position of object (u) and for the position of image (v) are Δu & Δv respectively. The error in the measurement of the focal length of the convex lens will be :

A) $f\left(\frac{\Delta v}{v}+\frac{\Delta u}{u}\right)$ B) $2f\left(\frac{\Delta u}{u}+\frac{\Delta v}{v}\right)$ C) $f^2\left(\frac{\Delta u}{u^2}+\frac{\Delta v}{v^2}\right)$ D) $\frac{\Delta v}{v}+\frac{\Delta u}{u}$

17. The co-ordinate of a particle moving in x-y plane are given by : $x = 2+4t;\ y = 3t+8t^2$. The motion of charged particle is :

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A) Uniform accelerated having motion along a parabolic path

B) Uniform accelerated having motion along a straight line

C) Non uniform accelerated motion

D) Uniform motion along a straight line

18. On Celsius scale the temperature of body increases by 40^0C. The increasing in temperature on Fahrenheit scale is:

A) 68°F B) 72°F C)75°F D)70°F

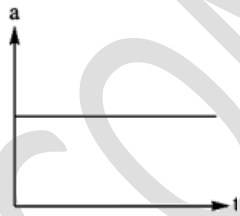
19. The electric field in an electromagnetic wave is given by

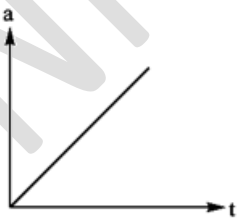
$\vec{E} = \hat{i}40\cos\omega\left(t - \frac{Z}{c}\right)NC^{-1}$. The magnetic field induction of this wave is (in SI unit) :

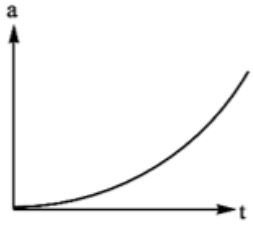
A) $\vec{B} = \hat{j}40\cos\omega\left(t - \frac{Z}{c}\right)$ B) $\vec{B} = \hat{j}\frac{40}{c}\cos\omega\left(t - \frac{Z}{c}\right)$

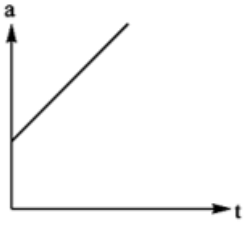
C) $\vec{B} = \hat{k}\frac{40}{c}\cos\omega\left(t - \frac{Z}{c}\right)$ D) $\vec{B} = \hat{i}\frac{40}{c}\cos\omega\left(t - \frac{Z}{c}\right)$

20. A wooden block is initially at rest on the ground, is pushed by a force which increases linearly with time t. Which of the following curve best describes acceleration of the block with time.

A) 

B) 

C) 

D) 

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. An elastic spring under tension of 3 N has a length a. Its length is b under tension 2 N. For its length (3a-2b), the value of tension will be _____N.

22. Two wavelength λ_1 and λ_2 are used in Young's double slit experiment. $\lambda_1 = 450nm$ and $\lambda_2 = 650nm$. The minimum order of fringe produced by λ_2 which overlaps with the fringe produced by λ_1 is n. The value of n is_____

23. Two forces $F_1 \& F_2$ are acting on a body. One force has magnitude thrice that of the other force and the resultant of the two forces is equal to the force of large magnitude. The angle between $F_1 \& F_2$ is $\cos^{-1}\left(\frac{1}{n}\right)$. The value of $|n|$ is_____.

24. A solid sphere and a hollow cylinder roll up without slipping on same inclined plane with same initial speed v. The sphere and the cylinder reaches upto maximum heights h_1 and h_2 , respectively, above the initial level. The ratio h_1: h_2 : is $\frac{n}{10}$. The value of n is_____.

25. A alternating current at any instant is given by $i(t) = \left[6 + \sqrt{56}\sin\left(2\pi t + \frac{\pi}{3}\right)\right] A$. The RMS value of current is……_____ (Ampere).

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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. The element which shows only one oxidation state other than its elemental form is:

A) Nickel B) Scandium C) Cobalt D) Titanium

27. One of the commonly used electrode is calomel electrode. Under which of the following categories, calomel electrode comes?

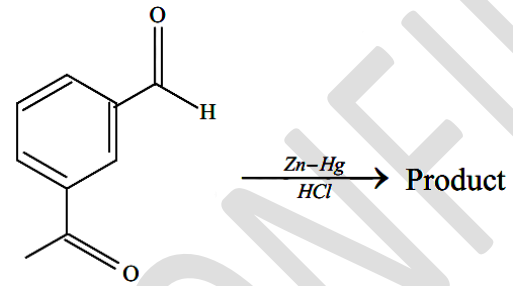
A) Metal ion - Metal electrodes

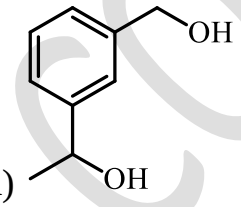
B) Oxidation - Reduction electrodes

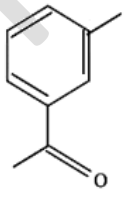
C) Metal Insoluble Salt - Anion electrodes

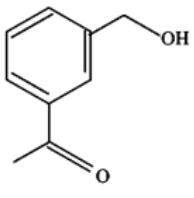
D) Gas Ion electrodes

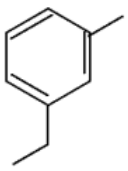
28. Identify the product in the following reaction:



A) 

B) 

C) 

D) 

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29. Which one of the following molecules has maximum dipole moment?

A) NF_3 B) PF_5 C)CH_4 D) NH_3

30. Number of elements from the following that can not form compounds with valencies Which Match with their respective group valencies is ————

$B, C, N, S, O, F, P, Al, Si$

A)7 B)3 C)5 D)6

31. Number of molecules/ions from the following in which the central atom is involved in sp^3 Hybridization is ———— $NO_3^-, BCl_3, ClO_2^-, ClO_3^-$

A)1 B)2 C)4 D)3

32. Number of complexes from the following with even number of unpaired "d" electrons is_

$\left[V(H_2O)_6\right]^{3+}, \left[Cr(H_2O)_6\right]^{2+}, \left[Fe(H_2O)_6\right]^{3+}, \left[Ni(H_2O)_6\right]^{3+}, \left[Cu(H_2O)_6\right]^{2+}$

[Given atomic numbers : V=23, Cr=24,Fe=26, Ni=28, Cu = 29]

A) 1 B) 4 C) 5 D) 2

33. Which among the following is incorrect statement?

A) The electromeric effect is temporary effect

B) The organic compound shows electromeric effect in the presence of the reagent only

C) Electromeric effect dominates over inductive effect

D) Hydrogen ion (H^+) shows negative electromeric effect

34. Which of the following nitrogen containing compound does not give Lassaigne's test?

A) Glycine B) Phenyl hydrazine

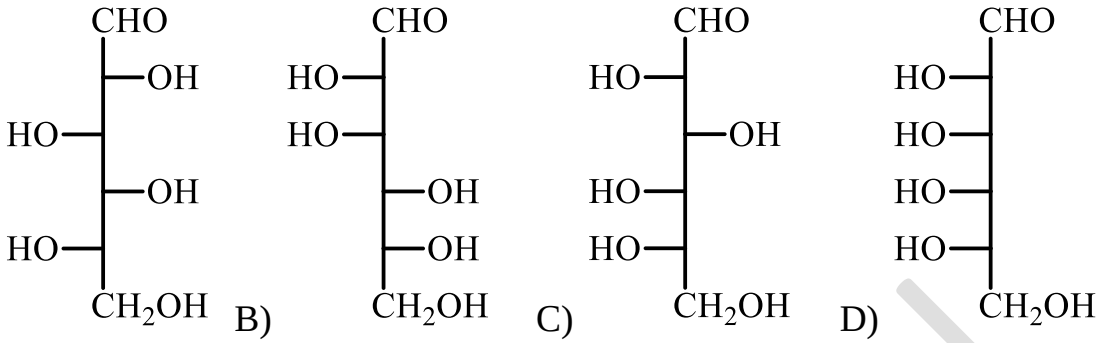
C) Urea D) Hydrazine

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35. Which of the following is the correct structure of L-Glucose?



36. **Statements 1:** Acidity of α -hydrogens of aldehydes and ketones is responsible for AldolReaction.

Statement II: Reaction between benzaldehyde and ethanal will not give Cross - Aldol product.

In the light of the above statements, choose the most appropriate answer from the Options given below:

A) Statement I is incorrect but Statement II is correct

B) Both Statement I and Statement II are correct

C) Both Statement I and Statement II are incorrect

D) Statement I is correct but Statement II is incorrect

37. The correct sequence of ligands in the order of decreasing field strength is:

A) $S^{2-} > OH^- > EDTA^{4-} > CO$ B) $OH^- > F^- > NH_3 > CN^-$

C) $NCS^- > EDTA^{4-} > CN^- > CO$ D) $CO > H_2O > F^- > S^{2-}$

38. What pressure (in bar nearly) of $H_{2(g)}$ would be required to make emf of hydrogen electrode zero in pure water at 25^0C?

A) 10^{-7} B) 0.5 C) 10^{-14} D) 1

39. Identify the correct set of reagents or reagents conditions 'X' and 'Y' in the Following set of transformation.

$$CH_3-CH_2-CH_2-Br \xrightarrow{'X'} Product \xrightarrow{'Y'} CH_3-\underset{\underset{Br}{|}}{CH}-CH_3$$

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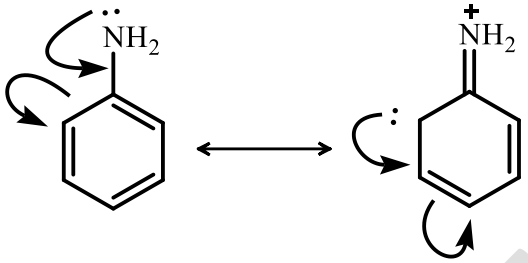
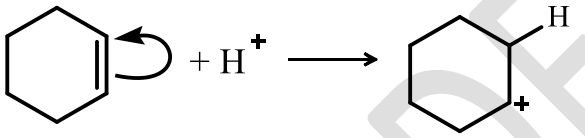
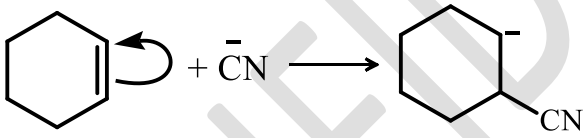
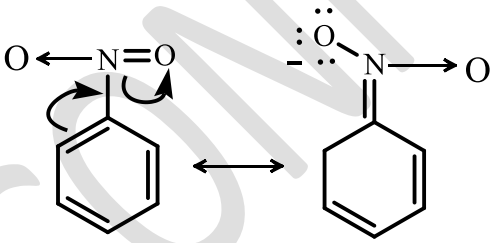
A) X = dil.aq.NaOH, 20^0C Y = $Br_2/CHCl_3$

B) X = dil.aq.NaOH, 20^0C Y = HBr / acetic acid

C) X = conc.alc.NaOH, 80^0C Y = $Br_2/CHCl_3$

D) X = conc.alc.NaOH, 80^0C Y = HBr/acetic acid

40. Match List I wish List II:

List – I **Mechanism steps**		**List – II** **Effect**
a)		I) $-E$ effect
b)		II) $-R$ effect
c)		III) $+E$ effect
d)		IV) $+R$ effect

A) $(a)-(III),(b)-(I),(c)-(II),(d)-(IV)$

B) $(a)-(II),(b)-(IV),(c)-(III),(d)-(I)$

C) $(a)-(I),(b)-(II),(c)-(IV),(d)-(III)$

D) $(a)-(IV),(b)-(III),(c)-(I),(d)-(II)$

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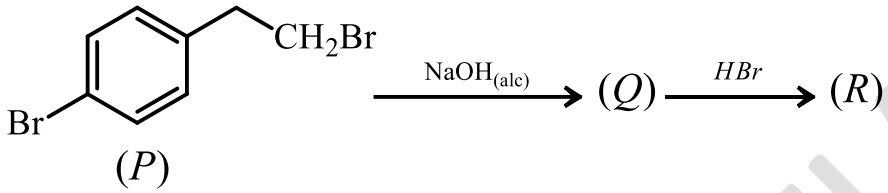
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41. The Molarity (M) of an aqueous solution containing 5.85 g of NaCl in 500mL water is nearly: (Given: Molar mass Na: 23 and Cl: 35.5 $gmol^{-1}$)

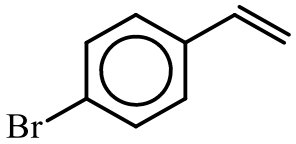
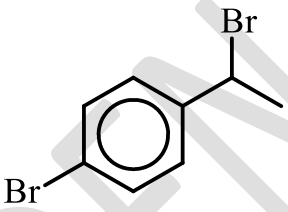
A) 20 B) 4 C) 2 D) 0.2

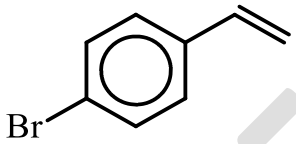
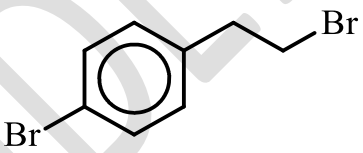
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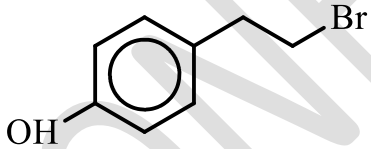
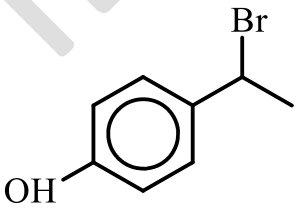


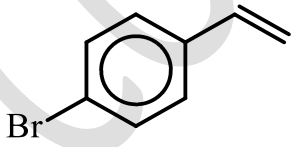
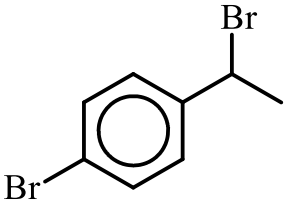
(P)

Identify (Q) and (R) and how are (P) and (R) related?

A)   ;position isomers

B)   ;chain isomers

C)   ;Derivative

A)   ; Functional isomers

43. The correct order of first ionization enthalpy values of the following elements is:

a) O b) N c) Be d) F e) B

Choose the correct answer from the options given below:

A) $c < e < a < b < d$ B) $b < d < c < e < a$

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C) $e < c < a < b < d$ D) $a < b < d < c < e$

44. What will be the decreasing order of basic strength of the following conjugate bases?

A) $OH^- > RO^- > CH_3COO^- > Cl^-$ B) $Cl^- > RO^- > OH^- > CH_3COO^-$

C) $RO^- > OH^- > CH_3COO^- > Cl^-$ D) $Cl^- > OH^- > RO^- > CH_3COO^-$

45. In the precipitation of the iron group (III) in qualitative analysis, ammonium chloride is added before adding ammonium hydroxide to

A) Increase concentration of NH_4^+ ions

B) Increase concentration of Cl^- ions

C) Decrease concentration of OH^- ions

D) Prevent interference by phosphate ions

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. The de-Broglie's wavelength of an electron in the 4th orbit is——— πa_0.

(where a_0 is Bohr's radius)

47. 2.5 g of a non-volatile, non-electrolyte is dissolved in 100 g of water at 25°C. The solution showed boiling point elevation by 2^0C. Assuming the solute concentration is negligible with respect to the solvent concentration, the vapor pressure of the resulting aqueous solution in mm of Hg at $100°C$ is (nearest integer) _____.

[Given: Molal boiling point elevation constant of water (K_b) = 0.52 K. kg mol^{-1}, 1atm Pressure =760 mm of Hg, Molar mass of water =18 g mol^{-1},

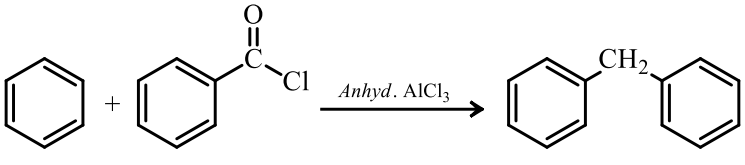
$\frac{20}{52} = 0.38, \frac{100}{18} = 5.55, \frac{0.38}{5.55} = 0.068$]

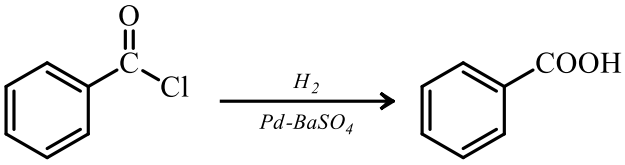
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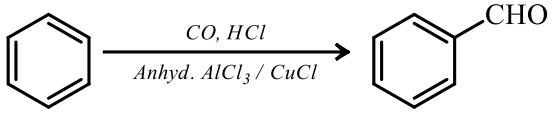
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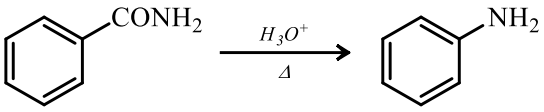
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48. The number of correct reaction(s) among the following is ————

i) 

ii) 

iii) 

iv) 

49. Consider the following reaction $MnO_2 + KOH + O_2 \rightarrow A + H_2O$ Product 'A' in neutral or acidic medium disproportionate to give products 'B' and 'C' along with water .The sum of spin –only magnetic moment values of B and C —— BM(nearest integer) (Given atomic number of Mn is 25)

50. Consider the following transformation involving first order elementary reaction in each step at constant temperature as shown below.

Some details of the above reactions as shown below

Step	Rate constant (sec^{-1})	Activation energy ($kJ\ mol^{-1}$)
1	k_1	300
2	k_2	200
3	k_3	Ea_3

If the overall rate constant of the above transformation (k) is given as $k = \frac{k_1 k_2}{k_3}$ and the overall activation energy (Ea) is 400 kJ mol^{-1}, then the value of Ea_3 is ———— kJ mol^{-1} (nearest integer)

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MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Consider three Urns A, B, C. A contains 7 red, 5 Black, B contains 5 red, 7 black and C contains 6 red, 6 black balls. One of the Urn is selected at random and a ball is drawn from it. If the ball drawn is black, then the probability that it is drawn from Urn A is

A) $\frac{5}{16}$ B) $\frac{5}{18}$ C) $\frac{7}{18}$ D) $\frac{4}{17}$

52. If $f(x)=\begin{cases} x-2, & 0<x\le 2 \\ -2, & -2\le x\le 0 \end{cases}$ and $h(x)=f(|x|)+|f(x)|$ then find $\int_{-2}^{2} h(x)dx$ is equal to

A) 2 B) 1 C) 6 D) 4

53. Let α *and* β be the sum and product of all non-zero solutions of the equation $(\bar{z})^2+|z|=0,\ z\in C,$ then $4(\alpha^2+\beta^2)$ is

A) 4 B) 2 C) 8 D) 6

54. If the solution y = y(x) of the differential equation $(x^4+2x^3+3x^2+2x+2)dy-(2x^2+2x+3)dx=0$ satisfies $y(-1)=-\frac{\pi}{4}$ then y(0) is

A) 0 B) $-\frac{\pi}{2}$ C) $\frac{\pi}{2}$ D) $\frac{\pi}{4}$

55. The sum of maximum and minimum values of the function $f(x)=\frac{2x^2-3x+8}{2x^2+3x+8}$ be $\frac{m}{n}$ where $\gcd(m,n)=1$ then $m+n=$

A) 217 B)182 C) 201 D) 195

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56. The sum of all rational numbers in the expansion of $\left(2^{1/5}+5^{1/3}\right)^{15}$ is equal to

A) 6131 B) 633 C) 931 D) 3133

57. Let $\alpha,\beta\in R$. Let mean and variance of 6 observations $-3,4,7,-6,\alpha,\beta$ be 2 and 23 respectively, the mean deviation about mean of these 6 observations is

A) $\frac{14}{3}$ B) $\frac{16}{3}$ C) $\frac{11}{3}$ D) $\frac{13}{3}$

58. Let $f:R\to R$ be a function given by $f(x)=\begin{cases}\frac{1-\cos 2x}{x^2} & ; \quad x<0\\ \alpha & ; \quad x=0,\\ \frac{\beta\sqrt{1-\cos x}}{x} & ; \quad x>0\end{cases}$ where

$\alpha,\beta\in R$. If f is continuous at x = 0. Then $\alpha^2+\beta^2$ equal to

A) 12 B) 6 C) 48 D) 3

59. The vertices of the triangle ABC are $A(-1,3),B(-2,2),$ and $C(3,-1)$. The new triangle is formed by shifting the sides of the triangle on by one-unit inwards. Then the equation of the side of the new triangle nearest to origin is:

A) $x+y+\left(2-\sqrt{2}\right)=0$ B) $x-y-\left(2+\sqrt{2}\right)=0$

C) $-x+y-\left(2-\sqrt{2}\right)=0$ D) $x+y-\left(2-\sqrt{2}\right)=0$

60. Let a unit vector, which makes angle of 60^0 with $2\hat{i}+2\hat{j}-\hat{k}$ and angle of 45^0 with $\hat{i}-\hat{k}$ be $\vec{C}$. Then $\vec{C}+\left(-\frac{1}{2}\hat{i}+\frac{1}{3\sqrt{2}}\hat{j}-\frac{\sqrt{2}}{3}\hat{k}\right)$ is

A) $\frac{\sqrt{2}}{3}\hat{i}+\frac{\sqrt{2}}{3}\hat{j}+\left(\frac{1}{2}+\frac{2\sqrt{2}}{3}\right)\hat{k}$ B) $\left(\frac{1}{\sqrt{3}}+\frac{1}{2}\right)\hat{i}+\left(\frac{1}{\sqrt{3}}-\frac{1}{3\sqrt{2}}\right)\hat{j}+\left(\frac{1}{\sqrt{3}}+\frac{\sqrt{2}}{3}\right)\hat{k}$

C) $\frac{\sqrt{2}}{3}\hat{i}+\frac{1}{3\sqrt{2}}\hat{j}-\left(\frac{1}{2}\right)\hat{k}$ D) $\frac{\sqrt{2}}{3}\hat{i}-\left(\frac{1}{2}\right)\hat{k}$

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61. A square is inscribed in a circle $x^2+y^2-10x-6y+30=0$. One side of the square is parallel to $y=x+3$. If (x_i, y_i) are the vertices of the square, then $\sum\left(x_i^2+y_i^2\right)$ is equal to :

A) 156 B) 152 C) 160 D) 148

62. If 2 and 6 are the roots of the equation $ax^2+bx+1=0$ then the quadratic equation whose roots are $\frac{1}{2a+b}$ *and* $\frac{1}{6a+b}$ is

A) $2x^2+11x+12=0$ B) $x^2+10x+16=0$

C) $x^2+8x+12=0$ D) $4x^2+14x+12=0$

63. If the system of equations

$x+\left(\sqrt{2}\sin\alpha\right)y+\left(\sqrt{2}\cos\alpha\right)z=0$

$x+\left(\cos\alpha\right)y+\left(\sin\alpha\right)z=0$

$x+\left(\sin\alpha\right)y-\left(\cos\alpha\right)z=0$ has a non-trivial solution, then $\alpha\in\left(0,\frac{\pi}{2}\right)$ is equal to :

A) $\frac{7\pi}{24}$ B) $\frac{3\pi}{4}$ C) $\frac{11\pi}{24}$ D) $\frac{5\pi}{24}$

64. Let $f(x)=x^5+2e^{\frac{x}{4}}$ for all $x\in R$. Consider a function g(x) such that (gof) (x) = x for all $x\in R$. Then the value of $8g'(2)$ is

A)16 B) 4 C) 8 D) 2

65. There are 5 points P_1, P_2, P_3, P_4, P_5 on the side AB, excluding A and B, of a triangle ABC. Similarly there are 6 points P_6, P_7….., P_{11} on the side BC and 7 points P_{12}, $P_{13,\ldots,}P_{18}$ on the side CA of the triangle. The number of triangles, that can be formed using the points

P_1, P_2,…..,P_{18} as vertices, is?

A) 776 B) 751 C) 771 D) 796

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66. Let $\alpha \in (0,\infty)$ and $A=\begin{bmatrix}1 & 2 & \alpha \\ 1 & 0 & 1 \\ 0 & 1 & 2\end{bmatrix}$. If $\det\left(adj\left(2A-A^T\right).adj\left(A-2A^T\right)\right)=2^8$, then $(\det(A))^2$ is equal to

A) 49 B) 1 C) 16 D) 36

67. Let the first three terms 2, p and q, with $q \neq 2$, of a G.P. be respectively the 7th, 8th and 13th terms of an A.P. If 5th term of the G. P is the nth term of the A.P. then n is equal to

A) 169 B)177 C) 163 D) 151

68. Let the point, on the line passing through the points $P(1,-2,3)$ and $Q(5,-4,7)$, farther from the origin and at a distance of 9 units from the point P, be (α,β,γ). Then $\alpha^2+\beta^2+\gamma^2$ is equal to

A) 160 B) 150 C) 155 D) 165

69. One of the points of intersection of the curves $y=1+3x-2x^2$ and $y=\frac{1}{x}$ is $\left(\frac{1}{2},2\right)$. Let the area of the region enclosed by these curves be $\frac{1}{24}\left(l\sqrt{5}+m\right)-n\log_e\left(1+\sqrt{5}\right)$, where $l,m,n \in N,$. Then $l+m+n$ is equal to

A) 29 B)31 C) 32 D) 30

70. If the domain of the function $\sin^{-1}\left(\frac{3x-22}{2x-19}\right)+\log_e\left(\frac{3x^2-8x+5}{x^2-3x-10}\right)$ is $(\alpha,\beta]$. Then $3\alpha+10\beta$ equals to

A) 97 B) 95 C) 98 D) 100

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Let A be a square matrix of order 2 such that |A| = 2 and the sum of its diagonal elements is –3. If the point (a,b), satisfying $A^2+aA+bI=O$, lies on the hyperbola, $\frac{x^2}{1}-\frac{y^2}{c^2}=1$ whose eccentricity is e and the length of the latus rectum is l, then $12\left(e^4+l^4\right)$ is equal to _____ .

72. If the shortest distance between the lines $\frac{x+2}{2}=\frac{y+3}{3}=\frac{z-5}{4}$ and $\frac{x-3}{1}=\frac{y-2}{-3}=\frac{z+4}{2}$ is $\frac{38}{3\sqrt{5}}k$, and $\int_0^k\left[x^2\right]dx=\alpha-\sqrt{\alpha}$, where [x] denote the greatest integer function, then $6\alpha^3$ is equal to ___

73. If $\lim\limits_{x\to1}\frac{(5x+1)^{1/3}-(x+5)^{1/3}}{(2x+3)^{1/2}-(x+4)^{1/2}}=\frac{m\sqrt{5}}{n(2n)^{2/3}}$, where gcd(m,n) = 1 , then 8m + 12n is equal to

74. $\int_0^{\pi/4}\left(\frac{\sin^2 x}{1+\sin x\cos x}\right)dx=\frac{1}{a}\log_e\left(\frac{a}{3}\right)+\frac{\pi}{b\sqrt{3}}$ where $a,b\in N$, then $a+b=$

75. Let the length of focal chord PQ of the parabola $y^2=12x$ be 15units and if the distance of PQ from the origin is p then $10p^2$ is equal to

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